Name = Muhammad Daniyal

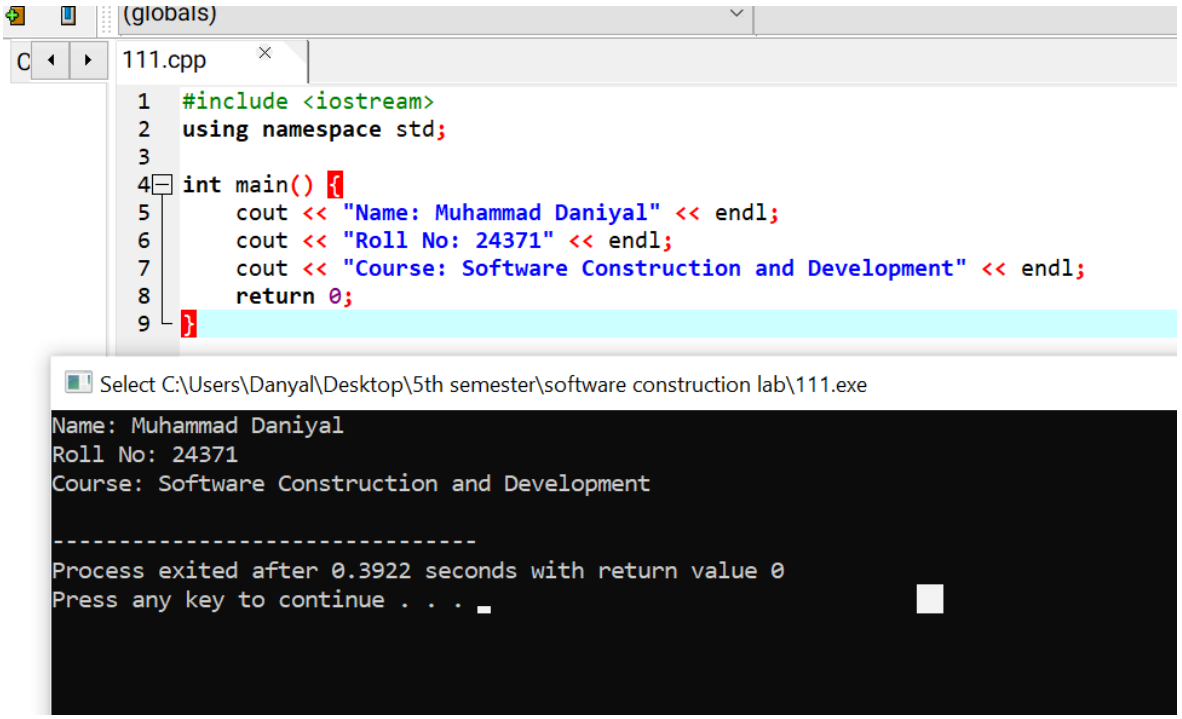
Registration no =24371

Software Construction and Development - Lab Assignment

1. Program to print Name, Registration Number, and Course Name

```
#include
<iostream> using
namespace std; int
main() {
  cout << "Name: Muhammad Daniyal" <<
  endl; cout << "Roll No: 24371" << endl;
  cout << "Course: Software Construction and
  Development" << endl; return 0;
}
```

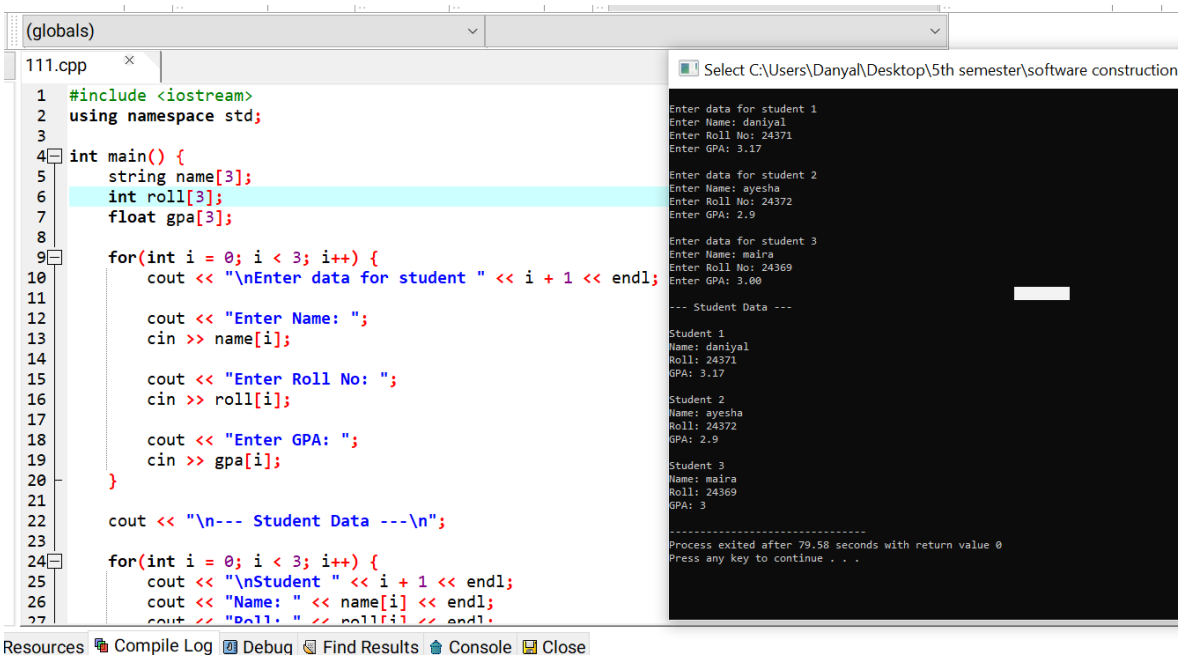
Output:



2. Input data of 3 students

```
#include
<iostream> using
namespace std; int
main() {
  string name[3]; int roll[3]; float
  gpa[3]; for(int i=0;i<3;i++){
    cin >> name[i] >> roll[i] >> gpa[i];
  }
  return 0;
}
```

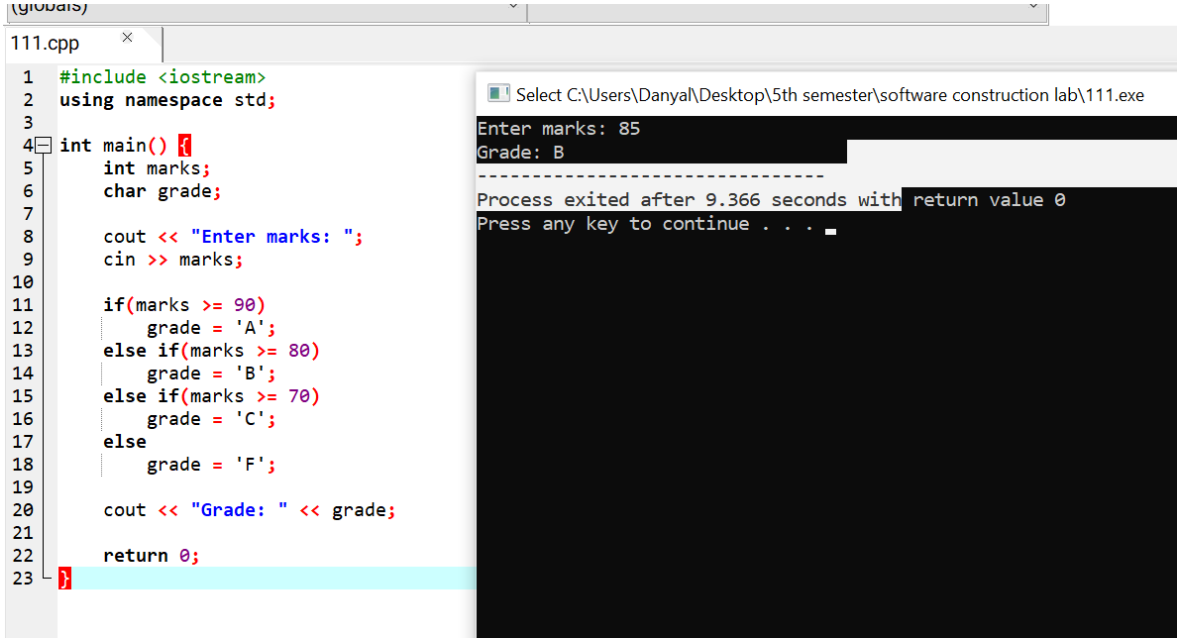
Output :



3. Grade Calculation using if-else

if(marks >= 90) grade = 'A';
else if(marks >= 80) grade = 'B'; else if(marks >= 70) grade = 'C'; else grade = 'F';

output:



4. Armstrong Number Function

```
bool isArmstrong(int
 num){ int sum=0,
 temp=num;
 while(temp!=0){
  int digit=temp%10;
  sum +=
  digit*digit*digit;
  temp/=10;
 }
 return sum==num;
}
```

Output:

```cpp
bool isArmstrong(int num) {
    int sum = 0, temp = num;

    while(temp != 0) {
        int digit = temp % 10;
        sum += digit * digit * digit;
        temp /= 10;
    }

    return sum == num;
}

int main() {
    int number;

    cout << "Enter a number: ";
    cin >> number;

    if(isArmstrong(number))
        cout << "It is an Armstrong number";
    else
        cout << "It is NOT an Armstrong number";

    return 0;
}
```

```
C:\Users\Danyal\Desktop\5th semester\software construction lab\111.exe
Enter a number: 153
It is an Armstrong number
--------------------------------
Process exited after 5.163 seconds with return value 0
Press any key to continue . . .
```

5. Find Max and Min in Array

int arr[10];
int max=arr[0],
min=arr[0]; for(int
i=1;i<10;i++){
if(arr[i]>max)
max=arr[i]; if(arr[i]<min)
min=arr[i];
}

Output:

```cpp
#include <iostream>
using namespace std;

int main() {
    int arr[10], max, min;

    for(int i = 0; i < 10; i++) {
        cin >> arr[i];
    }

    max = min = arr[0];

    for(int i = 1; i < 10; i++) {
        if(arr[i] > max)
            max = arr[i];
        if(arr[i] < min)
            min = arr[i];
    }

    cout << "Max: " << max << endl;
    cout << "Min: " << min << endl;

    return 0;
}
```

```
C:\Users\Danyal\Desktop\5th semester\software construction lab\111.exe
1
2
3
4
5
6
7
8
9
10
Max: 10
Min: 1

--------------------------------
Process exited after 15.85 seconds with return value 0
Press any key to continue . . .
```

6. Linear Search

```
for(int
 i=0;i<10;i++){
 if(arr[i]==key){
 cout <<
 "Found";
 }
}
```

Output:

(globals)

111.cpp

```cpp
#include <iostream>
using namespace std;

int main() {
    int arr[10], key;

    for(int i = 0; i < 10; i++) {
        cin >> arr[i];
    }

    cout << "Enter value to search: ";
    cin >> key;

    for(int i = 0; i < 10; i++) {
        if(arr[i] == key) {
            cout << "Value Found at index " << i;
            return 0;
        }
    }

    cout << "Value Not Found";
    return 0;
}
```

C:\Users\Danyal\Desktop\5th semester\software construction lab\111.exe

```
22
44
55
66
21
34
57
77
87
95
Enter value to search: 21
Value Found at index 4
--------------------------------
Process exited after 25.59 seconds with return value 0
Press any key to continue . . .
```

7. Bubble Sort

```
for(int
 i=0;i<10;i++){
 for(int
 j=0;j<9;j++){
 if(arr[j]>arr[j+1])
 {
    swap(arr[j], arr[j+1]);
  }
 }
}
```

Output:

111.cpp

```cpp
#include <iostream>
using namespace std;

int main() {
    int arr[10];

    for(int i = 0; i < 10; i++) {
        cin >> arr[i];
    }

    for(int i = 0; i < 10; i++) {
        for(int j = 0; j < 9; j++) {
            if(arr[j] > arr[j + 1]) {
                swap(arr[j], arr[j + 1]);
            }
        }
    }

    cout << "Sorted Array: ";
    for(int i = 0; i < 10; i++) {
        cout << arr[i] << " ";
    }

    return 0;
}
```

C:\Users\Danyal\Desktop\5th semester\software construction lab\111.exe

```
23
45
36
75
87
98
17
37
65
89
Sorted Array: 17 23 36 37 45 65 75 87 89 98
--------------------------------
Process exited after 20.33 seconds with return value 0
Press any key to continue . . .
```